

BHARANEE B

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SUMMARY

Computer Science graduate from VIT Chennai concurrently pursuing a B.S. in Data Science at IIT Madras. Hands-on experience architecting multi-agent GenAI systems (LangGraph, MCP), computer vision pipelines (PyTorch, YOLOv8), and scalable backend architectures (FastAPI, Redis, Celery). Comfortable owning systems end-to-end from data pipeline to production deployment.

TECHNICAL SKILLS

Languages: Python, Java, C/C++, SQL, JavaScript, HTML/CSS

Backend & Distributed: FastAPI, Flask, REST APIs, SSE Streaming, Pydantic v2, Vue.js, JWT & RBAC, Celery, Redis, SQLAlchemy

AI / ML & GenAI: LangGraph, Model Context Protocol (MCP), Hybrid RAG (Qdrant, BM25), Ragas, PyTorch, Hugging Face, Scikit-learn, OpenCV, YOLOv8, XGBoost, Pandas

Tools & Core CS: Git/GitHub, Docker, Postman, Linux, Data Structures & Algorithms, OOP, DBMS

EDUCATION

Vellore Institute of Technology Chennai

2022 – 2026

B.Tech. Computer Science Engineering • CGPA: 7.93

Indian Institute of Technology Madras

2022 – Present

B.S. Data Science and Applications (currently at Diploma level)

PROJECTS

Driver Risk Assessment System (DRAS) — Final-Year Capstone — Team of 3

Tech Stack: Python, PyTorch, OpenCV, YOLOv8, ResNet-34, Scikit-learn

- Engineered an edge-deployable multimodal pipeline fusing facial-expression recognition (ResNet-34), adaptive eye-aspect-ratio fatigue detection, head-pose estimation, and YOLOv8-based phone-distraction detection into one real-time driver risk score.
- Designed a late-fusion scoring model (70:30 visual-behavioral to affective weighting) across 4 independent perception streams, achieving **93.17% accuracy** on the NTHU-DDD drowsiness benchmark — beating the published 90.48% baseline and 3 alternative classifiers (MLP, KNN, SVM).
- Built continuous baseline-calibration logic to remove false positives from fixed-threshold fatigue detection, enabling real-time inference on standard consumer hardware with no per-driver training data.

Placement Portal Application — Full-Stack Campus Recruitment Platform | [GitHub](#)

Tech Stack: Python, Flask, Vue.js, Redis, Celery, JWT, SQLAlchemy

- Built a decoupled REST API (Flask Blueprints) with JWT-secured Role-Based Access Control powering distinct Admin, Company, and Student workflows across the full campus placement lifecycle.
- Introduced Redis-backed caching and Celery-driven asynchronous background jobs (e.g. CSV exports) to cut database load and eliminate blocking on high-traffic endpoints such as the job board.
- Shipped a responsive Vue.js 3 / Vite single-page app with Progressive Web App support for native-like installation across devices.

Enterprise Multi-Agent Intelligence Engine (FastMCP, LangGraph & Hybrid RAG) | [GitHub](#)

Tech Stack: Python, LangGraph, FastMCP (Model Context Protocol), Qdrant, BM25, FastAPI, Pydantic v2, SQLite, Ragas

- Architected an asynchronous multi-agent enterprise platform orchestrating Supervisor routing, self-correcting NL-to-SQL generation, and citation-grounded Hybrid RAG over financial transaction databases and credit underwriting policy documents.
- Engineered a two-stage retrieval pipeline fusing dense vector search (Qdrant 1536d) and sparse keyword retrieval (BM25 Okapi) via Reciprocal Rank Fusion (RRF, k=60) and Cross-Encoder reranking, achieving 100% Context Precision (Rank-1 retrieval).
- Decoupled tool services using FastMCP over Stdio/SSE transports with AST-level SQL validation (sqlparse) and URI read-only engine locks; streamed live tokens/thoughts via FastAPI SSE backed by Pydantic hallucination guardrails (117ms median latency).

Applied Machine Learning & Competitive Modeling — Kaggle Competitions

Tech Stack: Python, PyTorch, Hugging Face Transformers, XGBoost, LightGBM, CatBoost, Scikit-learn

- Benchmarked Linear Regression, Random Forest, XGBoost, LightGBM, and CatBoost for industrial equipment price prediction, implementing feature engineering, cross-validation, and hyperparameter tuning to minimize validation error.
- Built an end-to-end PyTorch semantic-segmentation pipeline (U-Net/DeepLab-style, Dice loss, IoU metric) for pixel-level classification and fine-tuned Hugging Face transformer models optimizing top-3 ranking via MAP@3.